Shuwei Zhao

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Education

The Chinese University of Hong Kong | 2023-09 - 2027-05

Bachelor of Engineering in Electronic Engineering | Hong Kong SAR

- CGPA: 3.911/4.000; Major GPA: 3.967/4.000. Expected graduation: May 2027.
- Engineering Elite Stream (2023-2026), a selective academic enrichment program for high-performing engineering students.

ETH Zurich | 2026-02 - 2026-08

Exchange Student in Electrical Engineering and Information Technology | Zurich, Switzerland

- Exchange period: February-August 2026; semester project: March-June 2026.
- Semester project at the Multi-Scale Robotics Lab under the supervision of Prof. Bradley Nelson; project grade 5.5/6.0.

Research Interests

Cable-driven mechanisms and actuation systems; wearable robotics and supernumerary robotic limbs; soft, compliant, and continuum robots; sensing, control, and experimental characterization.

Research Experience

Motorized Catheter Management Module for Industrial Inspection

Research Assistant | Xi'an Jiaotong University | 2026-07 - 2026-08
advisor, Prof. Qiji Ze.

- Designed a motorized translating-pulley catheter accumulator for insertion and retraction, with approximately twofold length compensation ($\Delta L \approx 2\Delta x$).
- Developed the assembly in Fusion 360 and SolidWorks using a 120-mm guide pulley, 608 bearings, an MGN12H carriage, a T8 lead-screw stage, and custom 3D-printed components.
- Implemented Python/Tkinter software, USB serial communication, Arduino UNO firmware, and TB6600 stepper control, including calibration, jogging, fixed-distance motion, adjustable speed and acceleration, travel limits, and watchdog protection.

Magnetically Steerable Micro-Camera Catheter for Bronchial-Tree Phantom Navigation

Research Assistant | The Chinese University of Hong Kong | 2026-06 - 2026-07
advisor, Prof. Hongliang Ren.

- Designed and fabricated a millimeter-scale catheter integrating an LED micro-camera, a 0.2-mm nitinol support, heat-shrink encapsulation, and a hard-magnetic Ecoflex cuff.
- Built insertion-force and steering-characterization setups and completed bronchial-tree phantom navigation experiments.
- Demonstrated multidirectional magnetic steering and forward-looking intraluminal imaging; led manuscript preparation as co-first author.

Soft Miniaturized Intraductal Imaging Catheter

Research Assistant (Semester Project) | ETH Zurich, Multi-Scale Robotics Lab | 2026-03 - 2026-06
supervisor, Prof. Bradley Nelson.

- Designed and fabricated flexible imaging catheter prototypes using nitinol support wires of 0.5 mm and 0.7 mm in diameter.
- Built a four-load-cell force platform with Arduino and Python data acquisition and compared prototypes with a 1.1-mm rigid reference.
- The reported analysis used 10 selected trials from 20 recorded manual trials per condition and included a 1.1-mm rigid reference; compared support and surface configurations in a simplified silicone setup.
- Demonstrated imaging in a silicone breast-duct phantom; semester project grade 5.5/6.0.

Research Experience (continued)

Pneumatic Soft Jumping Robot

Research Assistant | The Chinese University of Hong Kong | 2025-09 - present
advisor, Prof. Hongliang Ren.

- Designed and fabricated an Arduino-controlled pneumatic jumping robot using a double-layer spherical-cap silicone actuator.
- Developed geometric models and investigated the effects of actuator radius and opening angle on deformation and elastic-energy storage.
- Modeled actuator geometry and evaluated material stiffness and inflation-release cycles against jump height, frequency, and repeatability.

Bioinspired Kirigami Capsule Robot for Minimally Invasive Gastrointestinal Biopsy

Research Assistant | The Chinese University of Hong Kong | 2025-06 - 2025-08
advisor, Prof. Hongliang Ren.

- Developed A4988-based motor-control electronics and Arduino pulse-control software for a biopsy capsule measuring 17 mm in diameter and 22 mm in length.
- Built an angular calibration platform yielding approximately 18.1 degrees per pulse and R-squared of 0.99.
- Conducted ex vivo biopsy experiments and analyzed penetration, interaction-force, and tissue-yield data for the accepted ICRA 2026 paper.

Publications

[1] Shuwei Zhao†, Ruizhou Zhao†, Zhiqing Tang, Wenchao Yue, and Hongliang Ren. **A Magnetically Steerable Micro-Camera Catheter for Bronchial-Tree Phantom Navigation**. International Conference on Biomimetic Intelligence and Robotics (ICBIR), 2026. Accepted. Shuwei Zhao and Ruizhou Zhao contributed equally.

[2] Ruizhou Zhao†, Yichen Chu†, Shuwei Zhao, Wenchao Yue, Raymond Shing-Yan Tang, and Hongliang Ren. **Bioinspired Kirigami Capsule Robot for Minimally Invasive Gastrointestinal Biopsy**. IEEE International Conference on Robotics and Automation (ICRA), 2026. Accepted at IEEE ICRA 2026. Author manuscript available on arXiv: 2602.06207. Ruizhou Zhao and Yichen Chu contributed equally.

Awards

Song Qingling Scholarship (2023-2027) | Zhiyuan Education Fund; RMB 320,000 scholarship.

Dean's List (2023, 2024) | Faculty of Engineering, The Chinese University of Hong Kong; Awarded to students in the top 10 percent.

United College Head's List (2023, 2024) | United College, The Chinese University of Hong Kong; Recognition for outstanding academic performance.

Charles Kao Scholarship (2024, 2025) | The Chinese University of Hong Kong

Engineering Elite Stream Scholarship (2023, 2024) | The Chinese University of Hong Kong

Arthur and Louise May Memorial Scholarship (2023-2025) | The Chinese University of Hong Kong

Second Prize, Shandong Provincial Mathematics Competition (2022)

Skills

Mechanical Design and CAD: SolidWorks, Fusion 360, Onshape

Embedded Systems and Electronics: Arduino, motor control, sensor integration, schematic and PCB design, KiCad, EasyEDA Pro

Programming: C/C++, Python, MATLAB, LaTeX

Experimental and Fabrication: 3D printing, silicone and Ecoflex molding, nitinol catheter fabrication, force sensing, load-cell calibration, magnetic actuation, optical and infrared sensing, ex vivo tissue experiments

Selected Course Project

Autonomous Tracking and Grasping Robot | October 2024 - January 2025

Course Project | The Chinese University of Hong Kong | Advisor: Ms. Lai Yin NGAN.

- Built an autonomous mobile robot for line tracking, object acquisition, transportation, and placement within a four-minute competition task.
- Integrated multi-sensor tracking, a stepper-driven gripper, and high-friction contact surfaces.
- Developed inertial compensation, action sequences, and state-transition logic in C++.